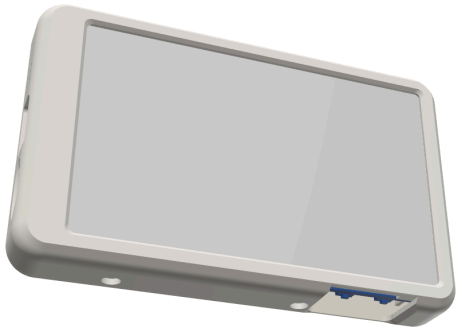
Case for the ESP32-S3-ePaper-3.97: assembly guide

What to buy, how to print it and what goes where



A two-part 3D-printed case (104.9 × 65.4 × 16.7 mm) for the Waveshare ESP32-S3-ePaper-3.97 board. The battery and the speaker sit side by side inside it, each in a bay made for **your own** battery and speaker, measured from your photos. You print a **front shell** and a **back cover**, put the board, battery and speaker in, and close it with 4 small screws. Nothing has to bend, so it works in any hard plastic; this version is printed in PLA.

What you need to buy

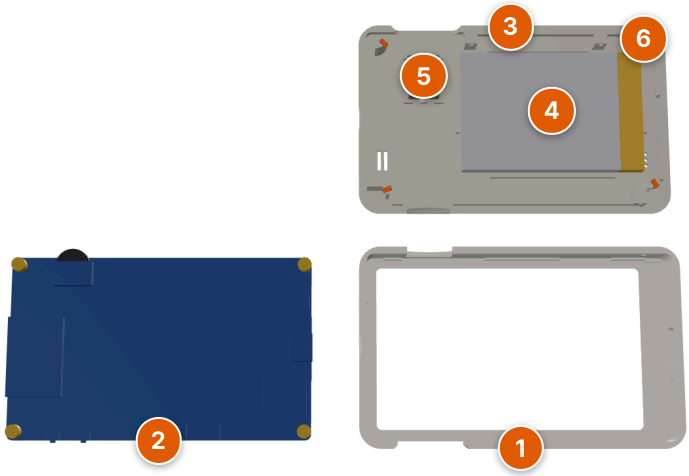
Item	Qty	Notes
Waveshare ESP32-S3-ePaper-3.97	1	The board itself (SKU 33552).
Waveshare 3.7 V 1500 mAh LiPo battery, model 454261	1	The bay is made for yours: measured 61.1 × 40.7 × 4.3 mm, bay 62.5 × 42.5 × 5 mm, so it goes in easily but can only shift about 0.9 mm.
The speaker that came with the board	1	The pocket is made for yours: measured 14.2 × 10.2 × 4.5 mm, cable about 10–11 cm. It has about 0.5 mm of room on every side.
M2 screws, 5–6 mm long, pan, cheese or button head	4	**Never longer than 7 mm**: the two bottom screws point at the battery. Head no more than 4.0 mm across and 1.4 mm tall, flat underneath (not countersunk, not washer-head). M2 machine screws are fine; they cut their own thread. Avoid 2.2 mm sheet-metal self-tappers.
Double-sided PE foam mounting tape, 1.0–1.2 mm thick	a few cm	Not thinner than 1.0 mm. Four small pieces hold the board tight.
Thin double-sided tape (0.1–0.2 mm)	a few cm	One piece under the battery, and a frame round the speaker's membrane (step 2).
PLA filament	about 26 g	For both printed parts (PrusaSlicer's estimate). PETG or PCTG also work with no change, and take heat better.

Tools

- A small screwdriver that fits your screws (usually Phillips PH0 or PH1, or a 1.5 mm hex key for hex-socket screws)
- Scissors or a craft knife for the foam tape
- Optional: calipers, to measure your board's thickness (see Tips)

The parts

Everything that goes into the case, laid out with the inside of each part facing you:



1 Front shell (printed). The board goes in here, display side down.

2 The board, seen from its back (components side).

3 Back cover (printed).

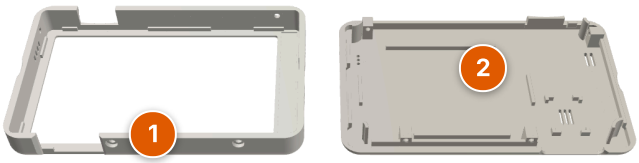
4 Battery in its bay (guides on all four sides).

5 Speaker in its pocket, over the three grille slots.

6 Foam pieces, one on each of the 4 corner rings.

Printing

- Material: **PLA**. It softens at about 55 °C: keep the case out of direct sun and hot cars. It is also more brittle than PETG, so pre-cut the screw threads (step 4) and don't over-tighten. PETG or PCTG work too, with no change.
- Layer height: **0.2 mm or finer** (0.1–0.2 mm). Don't use 0.3 mm.
- 3 perimeters (walls), 15–20 % infill.
- **No supports needed.**
- Front shell: print it **display side down**. Back cover: the file is already turned over; print it **outside face down**.



1 Front shell: display side down on the bed.

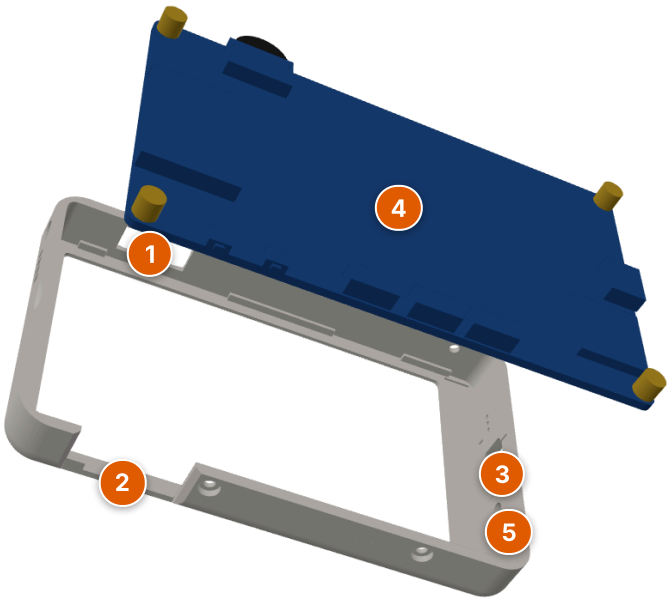
2 Back cover: outside face down on the bed.

> **Print the front shell first** (about 9 cm³) and check that the board drops in before you print the back cover. If you use ASA or ABS, turn on your slicer's XY shrinkage compensation for that filament: those plastics shrink enough to make the board too tight.

Assembly

1 Put the board into the front shell

Lay the front shell display side down. Put the board in **display side down**. The dial drops into the notch on the top edge, the PWR/BOOT buttons sit in the opening in the bottom edge, and the USB-C and microSD ports line up with their openings in the end. The picture is seen from the back.

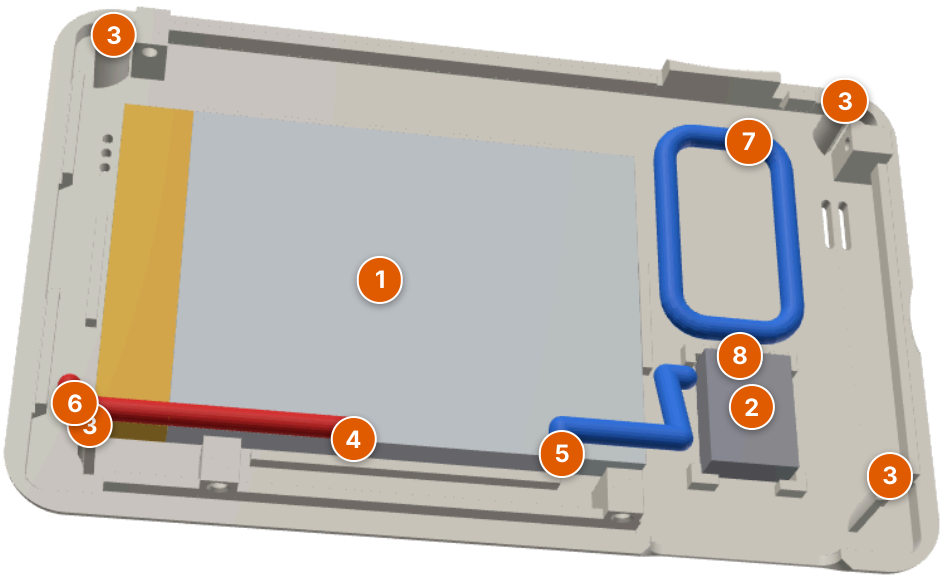


1. Dial notch (top edge).
2. PWR / BOOT opening (bottom edge).
3. USB-C opening.
4. The board, display side down.
5. microSD slot.

2 Battery, speaker and wires in the back cover

Hold the back cover inside up with the dial tab at the top, as in the picture. **Battery:** tape it into its bay with the **taped end (with the wire) toward the USB-C end**; if the wire comes out of the top corner, turn the cell over. **Speaker:** into its pocket **membrane side down** (white membrane in a black ring, facing the slots), **white peg toward the top edge**, so its cable faces the battery. Stick it with a frame of thin tape round the membrane, leaving the membrane uncovered. Plug the battery into **BAT** and the speaker into **SPK** (on the board's back, along its bottom edge) and run both wires along the bottom edge (about 38 mm and 30 mm). Coil the spare speaker cable in one loop above the speaker and tape it down, away from the right-hand end where the antenna is.

> **Check the connector:** BAT, RTC and SPK sit next to each other. The battery goes into **BAT** only. Make sure no wire lies across a screw hole or gets pinched when the cover closes.



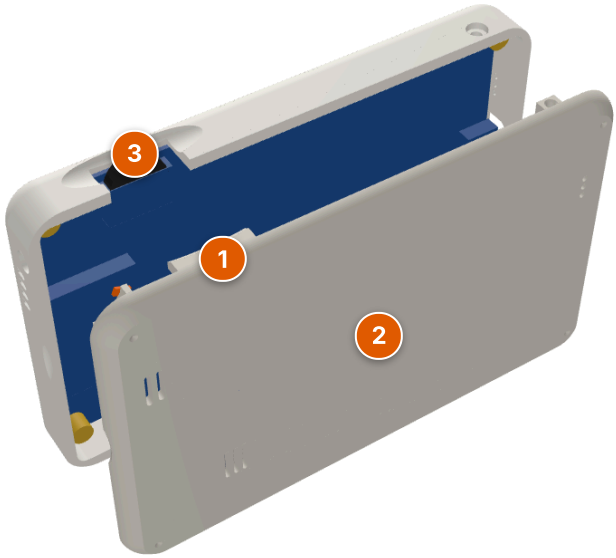
1. Battery in its bay (wire end toward the USB-C end).
2. Speaker, membrane side down over the slots.
3. Foam pieces on the 4 corner rings (step 3).
4. Battery plug → BAT connector.
5. Speaker plug → SPK connector.
6. The battery wire leaves the bay here.
7. Spare speaker cable: one loop, taped down.
8. The speaker's white peg points to the top edge.

3 Foam pieces on the 4 corner rings

Cut 4 pieces of the foam tape, about 1.5 × 3 mm. Stick one onto each of the 4 small curved rings in the back cover's corners (marked 3 in the picture above), then **peel off the top liner too**, so the foam also sticks to the board when the case is closed. The rings stop 0.6 mm short of the board, so the foam is always squeezed: it presses the board onto the front shell and holds it in every direction, so it can't rattle.

4 Close the case

First pre-cut the threads: drive each of the 4 screws about 4 mm into the hole in its block on the back cover, from the side that will face the wall, then back it out. Then, with the wires plugged in, **turn the back cover over like the page of a book along the bottom edge**, where the wires run, so they aren't pulled. Hold it over the front shell with its **dial tab** above the slot over the dial, then lower it **straight down**. Press until the seam closes all round. The foam gives a light, springy resistance.

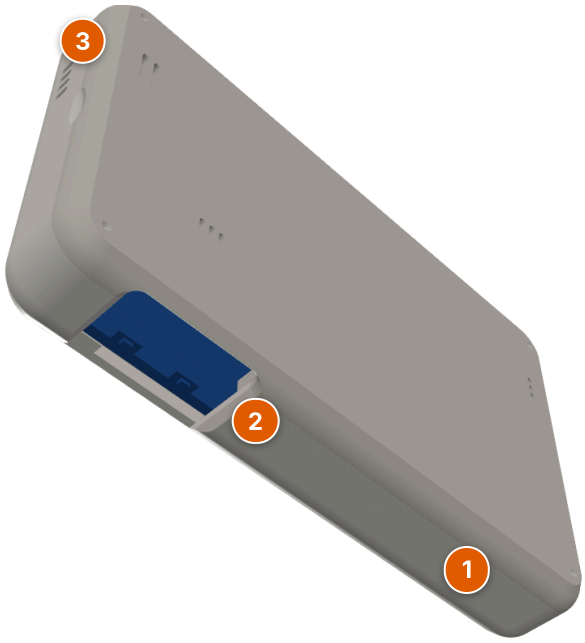


1. Dial tab: it goes into the slot above the dial.
2. Back cover: lower it straight down.
3. The dial in its notch.

5 Drive in the 4 screws

Hold the cover closed and drive in the 4 screws (their threads are already cut, step 4): **two in the bottom edge**, **one in the right-hand end** and **one in the top edge near the top-left corner** (left and right as seen from the display side). Stop as soon as each head **just touches the bottom of its recess**.

> Tighten them top and bottom alternately. **Don't over-tighten**: the thread is in plastic and can strip, and too much force pulls the cover sideways. **Never use screws longer than 7 mm.**



1. Bottom edge, screw 1.
2. Bottom edge, screw 2.
3. Right-hand end.



4. Top edge, near the top-left corner.

Opening and reassembling

- Take out the 4 screws and lift the back cover. **The board comes out with the cover**, because the foam is stuck to both.
- To separate the board from the cover, peel the foam off the board. Foam tape usually tears when peeled, so **use fresh pieces** when you put it back together.
- When you put a screw back in, first **turn it backwards** until it drops into its old thread, then tighten it, so it doesn't cut a new thread.

Tips

- The PWR and BOOT buttons stay fully uncovered in the bottom edge; press them straight on or slightly tilted.
- To eject the microSD card, press it in with a fingernail or a pen tip: it sits about 4 mm inside the edge.
- The board's thickness is assumed to be 1.6 mm. If yours is very different, measure it with calipers and set PCB_T in `case.py`, then rebuild the files (see README).
- The battery and speaker sizes were read off the ruler in your photos (about ±0.3 mm); the board's component heights are estimates. A test fit of the front shell is the real check.

Files: `stl/front_shell_battery.stl` and `stl/back_cover_battery.stl`. Details and every assumption are in README.md; all checks are in DEBUG_LOG.md.